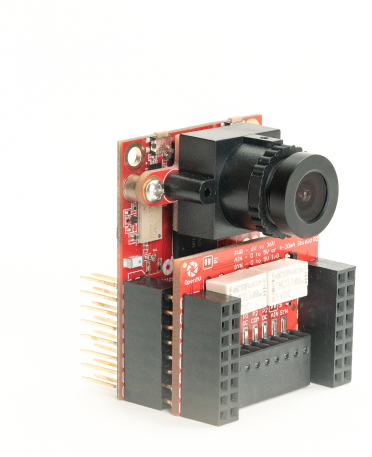
OpenMV® Relay Shield

Datasheet

SKU: OMV-SHIELD-RELAY

Description

The Relay Shield allows your OpenMV Cam to directly control two high-power DC and AC loads up to 60 W each. It features two independent relays controlled by P0 and P2 (or P1). This allows combining the Relay Shield with CAN/RS232 or RS422/RS485 shields which use P1 (or P2) and P3. Relays can be set to normally closed or normally open (default) independently via solder-bridge jumpers. Includes 6\u201336 V wide-range input, robust ADC, and trigger I/O.



Target Areas

High-power load switching, AC/DC control, lighting control, HVAC actuation, industrial automation, solenoid control.

Key Features

- **Two 60 W relays** — Two independent relays for high-power DC and AC loads up to 60 W each.
- **NO/NC configurable** — Each relay independently configurable as normally open (default) or normally closed via solder bridge.
- **Shield stacking** — Compatible with CAN/RS232 or RS422/RS485 shields via pin selection (P0/P1/P2).
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.
- **Robust ADC input** — 0–5 V on P6 with ±36 V overvoltage protection. Optional 4–20 mA mode.
- **Robust trigger I/O** — 0–5 V on P10 via SYN pin.

Contents

1 Features

- 1.1 Relay Outputs
- 1.2 Power Input
- 1.3 ADC Input
- 1.4 Digital I/O
- 1.5 Compatibility

2 General Info

3 Company Information

4 Reference Documentation

5 Revision History

1 Features

1.1 Relay Outputs

Component	Details
Relay 1	Controlled by P0. Up to 60 W DC or AC load. Configurable as normally open (default) or normally closed via solder-bridge jumper. Disconnectable.
Relay 2	Controlled by P2 (or P1 via solder bridge). Up to 60 W DC or AC load. Same NO/NC configuration. Disconnectable.

1.2 Power Input

Component	Details
Wide-Range Input	6–36 V with reverse voltage and transient surge protection. Outputs 5.4 V on VIN via OR'ing diode.

1.3 ADC Input

Component	Details
AIN	Robust 0–5 V ADC input on P6. Overvoltage protection to ±36 V. Optional 4–20 mA mode. Disconnectable.

1.4 Digital I/O

Component	Details
SYN / P10	Robust 0–5 V digital I/O. Tolerant to -12 V, shorts to GND and 12 V. Configurable input/output. Disconnectable.

1.5 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	11 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/relay-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.